A MINIMAL QUASI-COMPACT SUBSET MAPPING ONTO A QUASI-COMPACT SPACE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The statement that a quasi-compact subset of a compact space is closed. Invoked at the start of the counted proof (source line 5511). Identifies quasi-compact subsets with closed subsets.
- The statement that an intersection of nonempty closed subsets of a quasi-compact space is nonempty. Invoked at the end of the counted proof (source lines 5524-5526). Shows the intersection of a totally ordered family is again in the family.

Source reference. Original lemma lemma-closed-in-compact: sources/stacks-topology/pinned-original-source.tex, source lines 1923–1923. Navigation pointer only: the counted text refers to the Stacks item lemma-closed-in-compact, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-intersection-closed-in-quasi-compact: sources/stacks-topology/pinned-original-source.tex, source lines 1941–1941. Navigation pointer only: the counted text refers to the Stacks item lemma-intersection-closed-in-quasi-compact, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let $f : X \to Y$ be a continuous surjective map of Hausdorff quasi-compact topological spaces. There exists a quasi-compact subset $E \subset X$ such that $f(E) = Y$ but $f(E') \neq Y$ for all proper closed subsets $E' \subset E$.*

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Proof. We will use without further mention that the quasi-compact subsets of X are exactly the closed subsets (Lemma lemma-closed-in-compact). Consider the collection $\mathcal{E}$ of all quasi-compact subsets $E \subset X$ with $f(E) = Y$ ordered by inclusion. We will use Zorn's lemma to show that $\mathcal{E}$ has a minimal element. To do this it suffices to show that given a totally ordered family E_λ of elements of $\mathcal{E}$ the intersection $\bigcap E_\lambda$ is an element of $\mathcal{E}$. It is quasi-compact as it is closed. For every $y \in Y$ the sets $E_\lambda \cap f^{-1}(\{y\})$ are nonempty and closed, hence the intersection $\bigcap E_\lambda \cap f^{-1}(\{y\}) = \bigcap (E_\lambda \cap f^{-1}(\{y\}))$ is nonempty by Lemma lemma-intersection-closed-in-quasi-compact. This finishes the proof. $\square$